

Introduction to Groups

Lesson 2.1: Groups

Learning Outcomes: At the end of the lesson, the students are able to

- state the group axioms precisely;
 - identify whether a given set is a group under a given binary operations; and
 - prove the elementary properties of a group;
-

The Group Axioms

Let us start with a simple algebra problem of solving the linear equation

$$a + x = b,$$

where $a, b \in \mathbb{Z}$.

We first recognize that a has an additive inverse in $\mathbb{Z}$, namely $-a$, and add it to both sides of the equation.

$$(-a) + (a + x) = (-a) + b.$$

Since $+$ is associative, we may express the left-hand side of the equation as

$$[(-a) + a] + x = (-a) + b$$

$$0 + x = (-a) + b.$$

The fact that 0 is the identity element of $\mathbb{Z}$ under $+$ yields the conclusion that

$$x = (-a) + b$$

successfully solving for the value of x .

If we take note of the steps in this argument, we made use of the following properties:

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1. The operation $+$ is associative.
2. The set $\mathbb{Z}$ has an identity element 0.
3. The element $a \in \mathbb{Z}$ has an additive inverse, namely $-a$.

From this particular scenario, we see that in order for linear equations to be solvable in a given *algebraic structure*, these three properties must be imposed.

Definition 2.1.1 (Definition of a Group)

A **group** is an ordered pair $(G, *)$, where G is any nonempty set of objects, and $*$ is a binary operation on G , which satisfies the following **group axioms**:

G1. The binary operation $*$ is associative on G , that is,

$$(a * b) * c = a * (b * c), \text{ for all } a, b, c \in G.$$

G2. The set G has an identity element e under $*$, that is,

$$\text{there exists } e \in G \text{ such that } a * e = a = e * a, \text{ for all } a \in G.$$

G3. Every element of G is invertible, that is,

$$\text{for all } a \in G, \text{ there exists } a' \in G \text{ such that } a * a' = e = a' * a.$$

The **order** of the group $(G, *)$ is defined as the cardinality of the set G . A group with finite order is called a **finite group**. Otherwise, we call it an **infinite group**.

Although the commutative property is not part of the group axioms, it is possible that the given binary operation is commutative.

Definition 2.1.2 A group $(G, *)$ is **abelian** if $*$ is a commutative binary operation on G . Otherwise, we say that G is **non-abelian**.

Remark 2.1.3 By **Theorem 1.1.19**, every group $(G, *)$ has a unique identity element and by **Theorem 1.1.26**, every element of G has a unique inverse.

We now list down some examples of groups.



Example 2.1.4 The ordered pair $(\mathbb{Z}, +)$ is an abelian group, where $+$ is ordinary number addition. In a similar way, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$ are also abelian groups. Note that $(\mathbb{N}, +)$ is *not* a group since it doesn't have an identity element.

Let $\cdot$ denote integer multiplication. Then $(\mathbb{Z}, \cdot)$ is *not* a group. Although integer multiplication is associative and 1 is the identity element of $\mathbb{Z}$ under multiplication, not every element of $\mathbb{Z}$ is invertible under multiplication. For example, the integer 2 is not invertible under multiplication since $2 \cdot n \neq 1$, for any integer n .

Example 2.1.5 Let $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ denote the set of all *nonzero* rational numbers. One checks that $(\mathbb{Q}^\times, \cdot)$ is an abelian group, where $\cdot$ denotes the standard multiplication in $\mathbb{Q}$:

$$\frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd},$$

where $a, b, c, d \in \mathbb{Z}$ and $b, d \neq 0$. The identity element of this group is the rational number 1, and the inverse of a nonzero rational number is given by the formula

$$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}.$$

Example 2.1.6 For any integer $n > 1$, the pair $(\mathbb{Z}_n, +)$ is an abelian group, where $+$ denotes addition modulo n . As a particular example, the Cayley table for $(\mathbb{Z}_4, +)$ is given in [Table 6](#).

Example 2.1.7 Consider the set $V_4 = \{e, a, b, c\}$, where e , a , b , and c are maps of the form $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by the following formulas for all $(x, y) \in \mathbb{R}^2$:

$$\begin{aligned} e(x, y) &= (x, y) & a(x, y) &= (-x, y) \\ b(x, y) &= (x, -y) & c(x, y) &= (-x, -y) \end{aligned}$$

Let $\circ$ denote composition of functions. We claim that $(V_4, \circ)$ is a group. For instance, if $(x, y) \in \mathbb{R}^2$, we see that

- $(a \circ c)(x, y) = a(c(x, y)) = a(-x, -y) = (x, -y) = b(x, y)$; and
- $(b \circ a)(x, y) = b(a(x, y)) = b(-x, y) = (-x, -y) = c(x, y)$.

Thus, $a \circ c = b$ and $b \circ a = c$. Computing all possible compositions of the elements of V , we can generate its Cayley table as in [Table 7](#). As we can see, the element e is the identity element and every element of V_4 is self-inverse.



+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table 6: Cayley Table for $(\mathbb{Z}_4, +)$

$\circ$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Table 7: Cayley Table for $(V_4, \circ)$

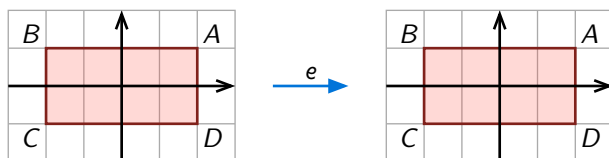
The group $(V_4, \circ)$ described in [Example 2.1.7](#) is known as the **Klein-4 group** named after the German mathematician **Felix Christian Klein (1849-1925)**, who laid a group theoretic foundation to the study of geometry known as the *Erlangen Program*. He used *symmetry groups* to classify geometric spaces giving a whole new approach to *non-euclidean geometries*. A *symmetry* of a geometric object X is any transformation of the space that makes X the image of itself.



Figure 3: Felix Klein

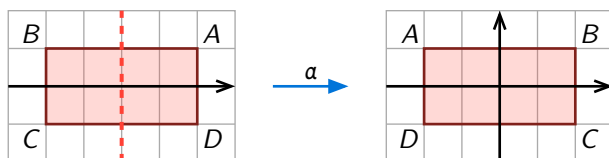
For example, the elements of the Klein-4 group may be realized as the symmetries of a rectangle in $\mathbb{R}^2$ whose vertices are at $A(p, q), B(-p, q), C(-p, -q), D(p, -q)$, where p and q are positive real numbers, with $p \neq q$.

1. the identity element $e : (x, y) \mapsto (x, y)$



e does nothing to the rectangle.

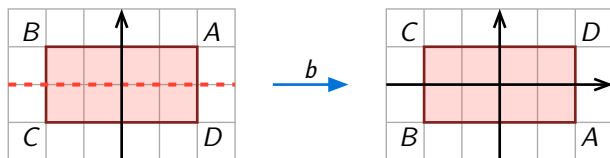
2. the element $a : (x, y) \mapsto (-x, y)$



a is a mirror reflection about the y -axis.

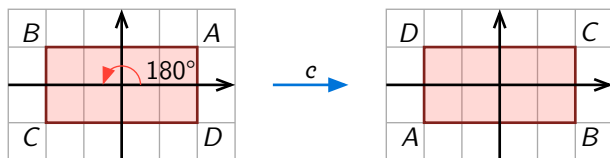


3. the element $b : (x, y) \mapsto (x, -y)$



b is a mirror reflection about the x -axis.

4. the element $c : (x, y) \mapsto (-x, -y)$



c is a 180° -rotation about the origin.

As the illustrations show, the group $(V_4, \circ)$ may be realized as the complete set of symmetries of a rectangle.

Example 2.1.8 Let X be an arbitrary set. Let $S(X)$ denote the set of all bijective functions of the form $\pi : X \rightarrow X$. We claim that $S(X)$ forms a group under composition of functions $\circ$.

- If $\alpha : X \rightarrow X$ and $\beta : X \rightarrow X$ are bijections, recall that $\alpha \circ \beta : X \rightarrow X$ is also a bijection. Hence, $\circ$ is a binary operation on $S(X)$.
- Composition of functions is generally associative.
- The identity map $\text{id}_X : X \rightarrow X$ given by $\text{id}_X(x) = x$, for all $x \in X$, is bijective so $\text{id}_X \in S(X)$. Additionally, for all $\alpha \in S(X)$, we have $\alpha \circ \text{id}_X = \alpha = \text{id}_X \circ \alpha$.
- Finally, if $\alpha \in S(X)$ is a bijection, recall that α is invertible. So a map $\alpha^{-1} : X \rightarrow X$ exists such that $\alpha \circ \alpha^{-1} = \text{id}_X = \alpha^{-1} \circ \alpha$.

Therefore, $(S(X), \circ)$ is a group.

Note that in general, $(S(X), \circ)$ is non-abelian. To see this, consider the case when $X = \{1, 2, 3\}$. The elements of $S(X)$ are the following bijections:



id_X	ρ_1	ρ_2	μ_1	μ_2	μ_3
$1 \mapsto 1$	$1 \mapsto 2$	$1 \mapsto 3$	$1 \mapsto 1$	$1 \mapsto 3$	$1 \mapsto 2$
$2 \mapsto 2$	$2 \mapsto 3$	$2 \mapsto 1$	$2 \mapsto 3$	$2 \mapsto 2$	$2 \mapsto 1$
$3 \mapsto 3$	$3 \mapsto 1$	$3 \mapsto 2$	$3 \mapsto 2$	$3 \mapsto 1$	$3 \mapsto 3$

We claim that $\rho_1 \circ \mu_1 \neq \mu_1 \circ \rho_1$. This is not hard to see since

$$(\rho_1 \circ \mu_1)(1) = \rho_1(\mu_1(1)) = \rho_1(1) = 2,$$

while

$$(\mu_1 \circ \rho_1)(1) = \mu_1(\rho_1(1)) = \mu_1(2) = 3.$$

Thus, $\rho_1 \circ \mu_1 \neq \mu_1 \circ \rho_1$. Therefore, $(S(X), \circ)$ is non-abelian.

This particular group is briefly denoted by S_3 since X has exactly three elements. The complete Cayley table for $(S_3, \circ)$ is shown in Table 8. The student is asked to verify the details of the computations.

$\circ$	id_X	ρ_1	ρ_2	μ_1	μ_2	μ_3
id_X	id_X	ρ_1	ρ_2	μ_1	μ_2	μ_3
ρ_1	ρ_1	ρ_2	id_X	μ_3	μ_1	μ_2
ρ_2	ρ_2	id_X	ρ_1	μ_2	μ_3	μ_1
μ_1	μ_1	μ_2	μ_3	id_X	ρ_1	ρ_2
μ_2	μ_2	μ_3	μ_1	ρ_2	id_X	ρ_1
μ_3	μ_3	μ_1	μ_2	ρ_1	ρ_2	id_X

Table 8: Cayley table for $(S_3, \circ)$

The group $(S(X), \circ)$ will be studied in depth in a dedicated lesson in the future.

If the context is clear, we sometimes identify a group by the set containing its elements by dropping the operation. For example, we may identify the group $(\mathbb{Z}, +)$ simply as the group $\mathbb{Z}$ if we agree in context that $\mathbb{Z}$ is the *additive group* of integers. The set $\mathbb{Z}$ is not a group under multiplication anyway. Hence, from this point, if it is not important to mention the binary operation of the group $(G, *)$, we shall sometimes refer to it briefly as the group G . In this case, the *product* of the elements $a, b \in G$ shall briefly be denoted by ab , if no confusion arises from using this notation. Note that the word *product* is simply used to refer to the result when a binary operation is performed and not necessarily refer to the result of a



multiplication. We call the manner of expressing products by placing elements side-by-side as *juxtaposition*. This is what we call as the *multiplicative notation*.

Under the multiplicative notation, we use the generic symbol e to refer to the identity element of a group and the inverse of an element a is denoted by a^{-1} , assuming that there is no conflict in notations. Of course, we cannot adapt this notation for additive groups such as $(\mathbb{Z}, +)$ or even in $(\mathbb{Z}_n, +)$, where the multiplicative notation is quite awkward. Unless otherwise stated, the multiplicative notation will be adapted in general discussions such as definitions and theorems.

We now introduce a way to construct new examples of groups out of the old ones. One way to do it is by considering *cartesian products*. To see this, let G and K be groups (possibly with different binary operations).

$$G \times K = \{(g, k) \mid g \in G \text{ and } k \in K\}.$$

On $G \times K$, define the *product* as follows:

$$(g_1, k_1) \cdot (g_2, k_2) := (g_1 g_2, k_1 k_2),$$

where $g_1 g_2$ is the product in G and $k_1 k_2$ as the product in K , relative to their respective binary operation. It is straightforward to check that $G \times K$ is a group under this operation. The students is asked to verify the details. The group $G \times K$ is called the **external direct product** of G and K .

Example 2.1.9 Consider the groups $(\mathbb{Z}_4, +)$ and $(S_3, \circ)$. In the group $\mathbb{Z}_4 \times S_3$,

$$(3, \rho_1) \cdot (2, \mu_2) = (3 + 2, \rho_1 \circ \mu_2) = (1, \mu_1)$$

since $3 + 2 = 1$ in $\mathbb{Z}_4$, under addition modulo 4, while $\rho_1 \circ \mu_2 = \mu_1$, under composition of functions in S_3 .

Note that we may extend the definition of external direct product to three or more groups. Let $G_1, G_2, \dots, G_n$ be groups. We define the external direct product

$$\prod_{i=1}^n G_i = G_1 \times G_2 \times \dots \times G_n := \{(g_1, g_2, \dots, g_n) \mid g_i \in G_i, \text{ for } i = 1, 2, \dots, n\}$$

as the group under the following binary operation:

$$(g_1, g_2, \dots, g_n) \cdot (h_1, h_2, \dots, h_n) := (g_1 h_1, g_2 h_2, \dots, g_n h_n),$$



where $g_i h_i \in G_i$ is the product with respect to the binary operation of the group G_i , $i = 1, 2, \dots, n$.

Elementary Properties of Groups

We now enumerate some important group properties which are quite handy to remember to prove more advanced results in group theory. We start with some properties of inverses.

Theorem 2.1.10 Let G be a group and let $a, b \in G$. Then

1. $(a^{-1})^{-1} = a$.
2. $(ab)^{-1} = b^{-1}a^{-1}$. **(Shoes and Socks Property)**

Proof. Let $a, b \in G$. Statement (1) follows from the *uniqueness of inverse* in a group and the fact that $aa^{-1} = e = a^{-1}a$. To prove statement (2), note that $ab \in G$, so ab is invertible by axiom (G3). By axioms (G1), (G2), and (G3), we have

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = ae a^{-1} = aa^{-1} = e;$$

and

$$(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = b^{-1}eb = b^{-1}b = e.$$

Again, by the *uniquess of inverse*, we have $(ab)^{-1} = b^{-1}a^{-1}$. □

Note that a common mistake is to assume that $(ab)^{-1} = a^{-1}b^{-1}$. In a non-abelian group, this is not true as the next example shows.

Example 2.1.11 Consider the group S_3 and the elements ρ_1 and μ_1 . Using [Table 8](#), we can see that

- $\rho_1 \circ \mu_1 = \mu_3$ so that $(\rho_1 \circ \mu_1)^{-1} = \mu_3^{-1} = \mu_3$.
- $\rho_1^{-1} = \rho_2$ and $\mu_1^{-1} = \mu_1$, so $\rho_1^{-1} \circ \mu_1^{-1} = \rho_2 \circ \mu_1 = \mu_2$.

These computations illustrate that

$$(\rho_1 \circ \mu_1)^{-1} \neq \rho_1^{-1} \circ \mu_1^{-1}.$$

The following remark immediately follows.

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Remark 2.1.12 In a group G , the equation $(ab)^{-1} = a^{-1}b^{-1}$ holds if and only if $ab = ba$.

Statement (2) of **Theorem 2.1.10** is also known as the *shoes and socks property* because of the order we wear and remove our shoes and socks, socks first and shoes next, unless you're crazy enough to wear your shoes even before you wear your socks! Removing them requires you to remove your shoes first, then the socks after.

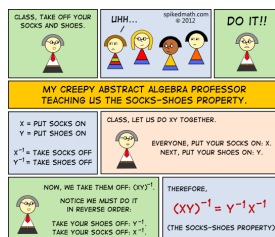


Figure 4: Shoes and Socks from Spiked Math

In a group G , the expression

$$a_1 a_2 \cdots a_n$$

is a well-defined product because of associative property (G1). It now makes sense to ask about the inverse of this product as a way to extend the shoes and socks property in a more general form. By applying mathematical induction, we have obtained the following.

Corollary 2.1.12.1 Let G be a group and $a_1, a_2, \dots, a_n \in G$. Then

$$(a_1 a_2 \cdots a_n)^{-1} = a_n^{-1} a_{n-1}^{-1} \cdots a_1^{-1}.$$

We next look at a group property that is useful in proving most theorems in group theory.

Theorem 2.1.13 (Group Cancellation Laws)

Let G be a group and let $a, b, c \in G$.

1. If $ab = ac$, then $b = c$. (left cancellation law)
2. If $ac = bc$, then $a = b$. (right cancellation law)

Proof. We only prove the left cancellation law and leave the proof of the right cancellation law to the student as an exercise.

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Assume that $ab = ac$. Then

$$b = eb = (a^{-1}a)b = a^{-1}(ab) = a^{-1}(ac) = (a^{-1}a)c = ec = c.$$

This shows that $b = c$. □

The group cancellation laws implies a remarkable property of finite groups. Let $G = \{a_1, a_2, \dots, a_n\}$ be a finite group with exactly n elements. Let $i, j, k \in \{1, 2, \dots, n\}$ such that $a_j \neq a_k$. By the contrapositive of the *left cancellation law*, we can conclude that $a_i a_j \neq a_i a_k$. This implies that the i th row of the Cayley table for G cannot contain the same element twice. Consequently, every element of G appears exactly once in every row of its Cayley table. In a similar fashion, using the *right cancellation law*, it can be deduced that every element of G appears exactly once in every column of its Cayley table. We shall refer to this property as the *sudoku property*, getting its name from a familiar puzzle.

Corollary 2.1.13.1 (Sudoku Property)

Let G be a finite group. Then every element of G appears exactly once in every row and every column of its Cayley table.

Example 2.1.14 Complete the partial Cayley table for the group $G = \{a, b, c, d\}$.

·	a	b	c	d
a	d		b	
b		b		
c				a
d			a	

Solution. Using the *sudoku property*, we first fill in the first row.

- Note that $ab = a$ or $ab = c$. If $ab = c$, then $ad = a$ contradicting the fact that $cd = a$. Thus, $ab \neq c$. Hence, $ab = a$. Consequently, $ad = c$. This completes the first row.

Next, we fill in the fourth column.

- Since $ad = c$ and $cd = a$, the either $bd = b$ or $bd = d$. However, $bb = b$ disqualifying the alternative $bd = b$. Thus, $bd = d$ so that $dd = b$.

Using the same logic, we fill in the second row.

- Note that $ba = a$ or $ba = c$. The option $ba = c$ forces $bc = a$, which violates the *sudoku property* because $dc = a$. Therefore, $ba = a$ and $bc = c$



We similarly apply the *sudoku property* to fill in the third row and fourth row to complete the table. The student is asked to fill in the details and the logical reasoning for each step. The complete Cayley table is shown below.

$\cdot$	a	b	c	d
a	d	a	b	c
b	a	b	c	d
c	b	c	d	a
d	c	d	a	b

We next prove a result which is closely related to the *sudoku property*. It asserts that given two elements a, b of a group G , then there are unique elements $x, y \in G$ such that $ax = b$ and $ya = b$.

Theorem 2.1.15 (Quasigroup Property)

Let G be a group and let $a, b \in G$.

1. The equation $ax = b$ has a unique solution in G for the unknown x .
2. The equation $ya = b$ has a unique solution in G for the unknown y .

Proof. We only prove statement (1) and ask the students to write a proof for (2).

Let $a, b \in G$ be given. We first prove the existence of a solution $x \in G$. For this, take $x = a^{-1}b \in G$. For this value of x , note that

$$ax = a(a^{-1}b) = (aa^{-1})b = eb = b.$$

Therefore, $x = a^{-1}b \in G$ is a solution to $ax = b$.

For the uniqueness, suppose that $x_1, x_2 \in G$ solves the equation $ax = b$. Then $ax_1 = b$ and $ax_2 = b$ implying that $ax_1 = ax_2$. By **Theorem 2.1.13** (1) (*left cancellation law*), it follows that $x_1 = x_2$ proving the uniqueness of such solution. $\square$

Example 2.1.16 Let $*$ be the binary operation on $\mathbb{Z}$ defined by

$$a * b := a + b - 2.$$

Show that $(\mathbb{Z}, *)$ is an abelian group and solve for the equation $1 * x * 4 = 10$.



Solution.

- Let $a, b \in \mathbb{Z}$. Then

$$a * b = a + b - 2 = b + a - 2 = b * a.$$

Therefore, $*$ is commutative.

- Let $a, b, c \in \mathbb{Z}$. Then

$$(a * b) * c = (a + b - 2) * c = (a + b - 2) + c - 2 = a + b + c - 4.$$

On the other hand,

$$a * (b * c) = a * (b + c - 2) = a + (b + c - 2) - 2 = a + b + c - 4.$$

Thus, $(a * b) * c = a * (b * c)$ proving that $*$ is associative.

- To check if $\mathbb{Z}$ has an identity element under $*$, we find an element $e \in \mathbb{Z}$ such that $a * e = a$, for all $a \in \mathbb{Z}$. This equation means that for all $a \in \mathbb{Z}$,

$$a + e - 2 = a \implies e = 2.$$

Indeed, if we take $e = 2$, then for any $a \in \mathbb{Z}$, we have

$$a * e = a + e - 2 = a + 2 - 2 = a.$$

Thus, $e = 2$ is the identity element of $\mathbb{Z}$ under $*$.

- Let $a \in \mathbb{Z}$ be given. We claim that there exists $a' \in \mathbb{Z}$ such that $a * a' = e = 2$. To see this, note that the equation $a * a' = 2$ means

$$a + a' - 2 = 2 \implies a' = 4 - a.$$

Indeed, if we take $a' := 4 - a$, then

$$a * a' = a + a' - 2 = a + (4 - a) - 2 = 2 = e.$$

Therefore, every $a \in \mathbb{Z}$ is invertible and its inverse is given by $a' = 4 - a$.

These arguments prove that $(\mathbb{Z}, *)$ is an abelian group. Finally, to solve for x in the equation $1 * x * 4 = 10$, we first let $y := 1 * x$. The given equation may be written as $y * 4 = 10$. By [Theorem 2.1.15](#)

$$y = 10 * 4' = 10 * (4 - 4) = 10 * 0 = 10 + 0 - 2 = 8.$$



Consequently, $1 * x = 8$. Again, by **Theorem 2.1.15**, we have

$$x = 1' * 8 = (4 - 1) * 8 = 3 * 8 = 3 + 8 - 2 = 9.$$

In elementary algebra, we have learned integer exponents to mean repeated multiplication. We adapt this convenient notation in group theory.

Definition 2.1.17 (Integer Exponents)

Let G be a group with identity e and let $a \in G$ be given. We define $a^0 := e$, $a^1 := a$, and

$$a^n := aa^{n-1},$$

for any integer $n > 1$. We also define

$$a^{-n} := (a^{-1})^n,$$

for any positive integer n .

Example 2.1.18 Let $a \in G$, where G is a group. Then

- $a^2 = aa^1 = aa$.
- $a^3 = aa^2 = a(aa) = aaa$.
- $a^{-3} = (a^{-1})^3 = a^{-1}a^{-1}a^{-1}$.

Note that because of associative property, grouping symbols may be removed without confusion.

- $a^3a^2 = (aaa)(aa) = aaaaaa = a^5$.
- $a^3a^{-2} = (aaa)(a^{-1}a^{-1}) = aa(\cancel{aa^{-1}}^e)a^{-1} = a(\cancel{aa^{-1}}^e) = a$

Warning

The *integer exponent* notation is used only when we are adapting the group multiplicative notation. Take note that exponents only indicate to perform the group operation repeatedly on a given element.

For additive groups like $(\mathbb{Z}, +)$, it is quite inappropriate to use the integer exponent notation as it means differently in this context. In this group, integer exponents



actually means *multiples*. For example, $n + n + n$ is actually equal to $3n$ instead of n^3 .

We also noticed from the previous example that certain rules of exponents hold in a group. We shall state them without proofs and allow the students to think about why each statement is true.

Theorem 2.1.19 (Law of Exponents)

Let G be a group and let $a, b \in G$. If $n, m \in \mathbb{Z}$, then

1. $a^n a^m = a^{n+m}$.
2. $(a^n)^m = a^{nm}$.
3. $(a^n)^{-1} = a^{-n}$.
4. $(ab)^n = a^n b^n$ if and only if $ab = ba$.

Example 2.1.20 Consider the group S_3 whose Cayley table is given in Table 8. Observe that

- $(\rho_1 \circ \mu_1)^2 = \mu_3^2 = \text{id}_X$; while
- $\rho_1^2 \circ \mu_1^2 = \rho_2 \circ \text{id}_X = \rho_2$.

As we can see, $(\rho_1 \circ \mu_1)^2 \neq \rho_1^2 \circ \mu_1^2$.

Definition 2.1.21 (Order of an Element)

Let G be a group with identity e . If there exists a positive integer n such that $a^n = e$, the smallest such n is called the **order** of a . Otherwise, we say that a has an **infinite order**. The order of a in G is denoted by the symbol $|a|_G$. In the case when a has an infinite order, we simply write $|a|_G = \infty$.

Immediate from Definition 2.1.21, we may deduce that $|a|_G = 1$ if and only if $a = e$.

Example 2.1.22 In the group $(\mathbb{Z}, +)$, every nonzero element has an infinite order since every positive multiple of a nonzero integer cannot be 0.

Example 2.1.23 In the group $(\mathbb{Z}_6, +)$, the order of each element is calculated as follows:



- $|0|_{\mathbb{Z}_6} = 1$, since 0 is the identity.
- $|1|_{\mathbb{Z}_6} = 6$, since $1 + 1 + 1 + 1 + 1 + 1 = 0$.
- $|2|_{\mathbb{Z}_6} = 3$, since $2 + 2 + 2 = 0$.
- $|3|_{\mathbb{Z}_6} = 2$, since $3 + 3 = 0$.
- $|4|_{\mathbb{Z}_6} = 3$, since $4 + 4 + 4 = 0$.
- $|5|_{\mathbb{Z}_6} = 6$, since $5 + 5 + 5 + 5 + 5 + 5 = 0$.

Example 2.1.24 Let $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ be the group of nonzero complex numbers under complex number multiplication.

Consider the element $i = \sqrt{-1}$.

Note that $|i|_{\mathbb{C}^\times} = 4$ since

- $i^2 = -1$;
- $i^3 = -i$; and
- $i^4 = 1$ (the identity).

On the other hand, the element $\sqrt{-4} = 2i$ has an infinite order since

$$(2i)^n = 2^n i^n \neq 1, \quad \text{for any positive integer } n.$$

Example 2.1.25 How many elements of order 2 are there in the group $\mathbb{Z}_2 \times \mathbb{Z}_4$?

Solution. We need to determine the elements $(a, b) \neq (0, 0)$ such that

$$(a, b) + (a, b) = (0, 0).$$

This condition implies $a + a = 0$ in $\mathbb{Z}_2$ and $b + b = 0$ in $\mathbb{Z}_4$.

- Note that for all $a \in \mathbb{Z}_2$, we have $a + a = 0$.
- On the other hand $b + b = 0$ in $\mathbb{Z}_4$ holds if and only if $b \in \{0, 2\}$.

Excluding $(0, 0)$, the elements satisfying these conditions are $(0, 2)$, $(1, 0)$, and $(1, 2)$.

Therefore, there are exactly 3 elements in $\mathbb{Z}_2 \times \mathbb{Z}_4$ which are of order 2.

